USN	1	R	V			E	C			

RV College of Engineering®
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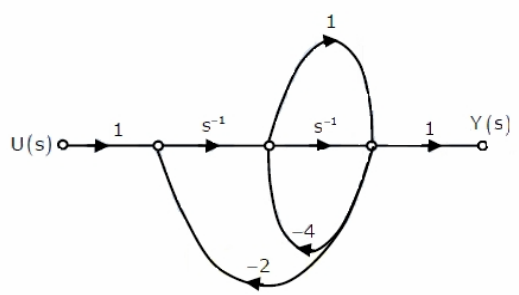
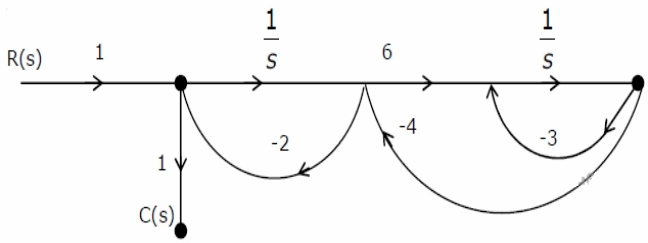
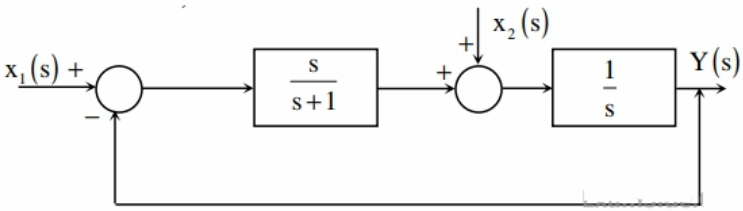
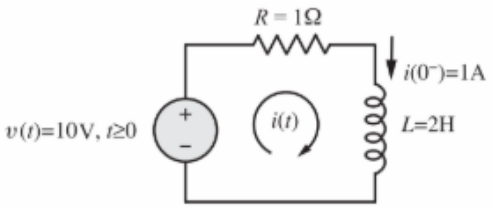
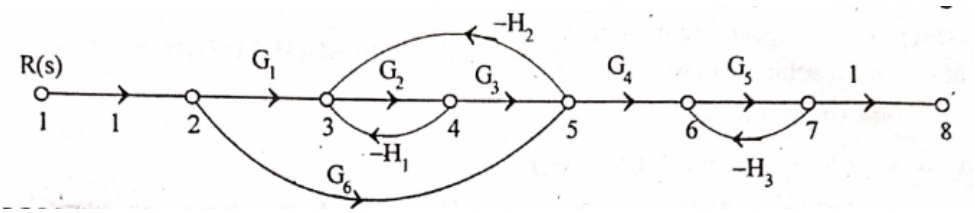
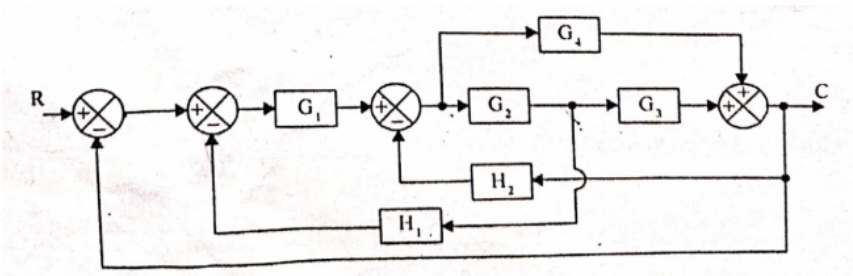
Bengaluru - 560059, Karnataka, India

NBA Accredited (UG - 6 Years
PG - 3 Years)

Department of Electronics & Communication Engineering

Date: 17/12/2025	**CIE -II**	**Max. Marks:60**
Semester:III	**UG**	**Duration:2 Hrs**
Course: Network Analysis and Control Engineering		**Code:EC235AT**

S No	Part-A Quiz	M	BT	CO
1.	The signal flow graph for a system is given in Fig-1 The transfer function Y(s) / U(s) for this system is _____  Fig -1	2	3	2
2.	The signal flow graph of a system is shown in fig-2. The transfer function C(s) / R(s) of the system is_____  Fig-2	2	2	2
3.	For the system given in Fig-3,the transfer function Y(s) / X_2(s) is _____  Fig-3	2	1	2
4.	For a second-order system with the closed-loop transfer function , the settling time for 2-percernt band, in seconds is_____	2	2	3
5.	Determine the initial and final value of	2	1	4
S No	**Part-B Test**	**M**	**BT**	**CO**

1	(a) Refer the circuit shown in Fig. 1. Find i(0) and i(∞) using initial and final value theorems.  Fig .1	6	3	1
	(b) Obtain the response of unity feedback system whose open loop T.F. is , and when the input is unit step.	6	3	3
2	Derive an expression for the unit step response of an under damped second order System.	10	2	3
3	Find the overall gain C(s)/R(s) for the signal-flow graph shown in Fig.3  Fig.3	10	3	4
4	Using block diagram reduction technique find closed loop transfer function of the system whose block diagram is shown in Fig. 4  Fig.4	10	3	3
5	Find all the time domain specifications for a unity feedback control system whose open loop transfer function is given by G(s) =	8	2	3

SCHEME & SOLUTIONS

S No	Solutions with Scheme	Marks
	Quiz	
1.	$\frac{s+1}{5s^2+6s+2}$	2
2.	$\frac{s(s+27)}{s^2+29s+6}$	2
3.	$\frac{s+1}{s(s+2)}$	2
4.	2 sec	2
5.	0, 0.553	2
	Test	
1(a)	*Applying KVL we get* $$i + 2\frac{di}{dt} = 10$$ Taking Laplace transform, on both the sides, we get $$I(s) + 2[sI(s) - i(0^-)] = \frac{10}{s}$$ $$\Rightarrow \quad I(s) + 2[sI(s) - 1] = \frac{10}{s}$$ $$\Rightarrow \quad I(s)[1 + 2s] = \frac{10}{s} + 2$$ $$\Rightarrow \quad I(s) = \frac{10}{s(1+2s)} + \frac{2}{1+2s} = \frac{10+2s}{s(1+2s)} = \frac{5+s}{s\left(s+\frac{1}{2}\right)}$$ According to initial value theorem, $$i(0) = \lim_{s\to\infty} sI(s) = \lim_{s\to\infty} s\frac{(s+5)}{s\left(s+\frac{1}{2}\right)} = \lim_{s\to\infty} \frac{1+\frac{5}{s}}{1+\frac{1}{2s}} = \mathbf{1}$$	

(b)	$i(\infty) = \lim_{s\to 0} sI(s)$ $= \lim_{s\to 0} \frac{s(s+5)}{s\left(s+\frac{1}{2}\right)} = \frac{5}{1/2} = \mathbf{10\ A}$ For unity feedback, $\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\frac{4}{s(s+5)}}{1+\frac{4}{s(s+5)}} = \frac{4}{s(s+5)+4} = \frac{4}{s^2+5s+4}$ So for a unit step $R(s) = \frac{1}{s}$, $C(s) = \frac{C(s)}{R(s)}R(s) = \frac{4}{s(s^2+5s+4)}$ Factor the quadratic: $s^2+5s+4 = (s+1)(s+4)$ So $C(s) = \frac{4}{s(s+1)(s+4)}$ $C(s) = \frac{1}{s} - \frac{4}{3}\frac{1}{s+1} + \frac{1}{3}\frac{1}{s+4}$ $\boxed{c(t) = 1 - \frac{4}{3}e^{-t} + \frac{1}{3}e^{-4t}, \qquad t \geq 0}$	
2.	Derivation	

3.

Forward path

From node 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8

$$P_1 = G_1G_2G_3G_4G_5$$

Individual loops

Loop L_1

3 → 4 → 3 (negative feedback through $-H_1$)

$$L_1 = -G_3H_1$$

Loop L_2

3 → 4 → 5 → 3 (via $-H_2$)

$$L_2 = -G_3G_4H_2$$

Loop L_3

6 → 7 → 6 (negative feedback via $-H_3$)

$$L_3 = -G_5H_3$$

Loop L_4

1 → 2 → 3 → 4 → 5 → 1 (via G_6)

$$L_4 = -G_1G_2G_3G_4G_6$$

$$L_1L_3 = (-G_3H_1)(-G_5H_3) = G_3G_5H_1H_3$$

$$L_2L_3 = (-G_3G_4H_2)(-G_5H_3) = G_3G_4G_5H_2H_3$$

Compute Δ

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_3 + L_2L_3)$$

Substitute:

$$\Delta = 1 - [(-G_3H_1) + (-G_3G_4H_2) + (-G_5H_3) + (-G_1G_2G_3G_4G_6)] + (G_3G_5H_1H_3 + G_3G_4G_5H_2H_3)$$

$$\Delta = 1 + G_3H_1 + G_3G_4H_2 + G_5H_3 + G_1G_2G_3G_4G_6 + G_3G_5H_1H_3 + G_3G_4G_5H_2H_3$$

Δ for forward path (touches all feedback loops)

$$\Delta_1 = 1$$

$$T(s) = \frac{C(s)}{R(s)} = \frac{P_1\Delta_1}{\Delta}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1G_2G_3G_4G_5}{1 + G_3H_1 + G_3G_4H_2 + G_5H_3 + G_1G_2G_3G_4G_6 + G_3G_5H_1H_3 + G_3G_4G_5H_2H_3}}$$

4.	$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1G_2G_3G_4 + G_1G_2G_4H_1 + G_2G_3G_4H_2 + G_2G_4H_1}{1 + G_1G_2H_1 + G_2G_3H_2 + G_2H_1}}$$	5 5

5.

$$\frac{C(s)}{R(s)} = \frac{G_{(s)}}{1 + G_{(s)} H_{(s)}}$$

But $H(s) = 1$ and $G(s) = \frac{25}{s(s+6)}$. Therefore

$$\frac{C(s)}{R(s)} = \frac{25}{s^2 + 6s + 25}$$

Therefore, $\omega_n^2 = 25$ or $\omega_n = 5$ rad/sec

$$2\zeta\,\omega_n = 6 \quad \text{or} \quad \zeta = \frac{6}{2 \times 5} = 0.6$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 5\sqrt{1 - 0.6^2} = 4 \text{ rad/sec}$$

$$\beta = \tan^{-1} \frac{\sqrt{1 - \zeta^2}}{\zeta} = \tan^{-1} \frac{\sqrt{1 - 0.6^2}}{0.6}$$

$$= 53.13^\circ = 0.92 \text{ rad}$$

(1) *Rise time*

$$t_r = \frac{\pi - \beta}{\omega_d} = \frac{3.14 - 0.92}{4} = 0.55 \text{ secs}$$

(2) *Peak time*

$$t_p = \frac{\pi}{\omega_d} = \frac{3.14}{5} = 0.785 \text{ secs}$$

(3 *Peak Overshoot*

$$M_p = e^{\frac{-0.6 \times \pi}{0.8}} = 9.5\%$$

(4) *Settling time (2%)*

$$t_s = \frac{4}{0.6 \times 5} = 1.33 \text{ secs}$$

4

6

SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Declaration by the Course handling faculties:

As ateaching faculty of the courseNetwork Analysis and Control Engineering with code:EC235AT. We hereby confirm that the question paper with Scheme and Solutions is thoroughly reviewed and we ensure that it adheres to thefollowing criteria:

Sl. No	Criteria	Yes/No
1	The question paper adequately covers the prescribed syllabus contents.	Y
2	The question paper is in line with the recommended pattern, taking intoconsideration the structure and format suitable for the Continuous Internal Evaluation.	Y
3	The question paper is designed to align with the Revised Bloom'sTaxonomy, encompassing various levels of cognitive skills such as remembering,understanding, applying, analysing, evaluating, and creating.	Y
4	The question paper is aligned with the defined course outcomes, ensuring that iteffectively assesses the knowledge and skills acquired during the course.	Y
5	Course handling faculty (as applicable) who are responsible for preparing thequestion paper, Scheme and Solutions have unanimously agreed to utilize this Questionpaper for conducting the Continuous Internal Evaluation.	Y
6	The Question paper, Scheme and complete Solutions have been submitted to the Testcoordinators within the designated time-frame to ensure the smooth conduction ofContinuous Internal Evaluations	Y

Course handling Faculties:

Name: **Signature**

1.

2.

3.

4.

To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	**Awarded**
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code,date, max marks, BT CO table, efficient use of paper (proper spacing,figures)	10	
4	No handwritten data or diagrams, and uniform fonts throughout	10	
5	Scheme &complete Solutions in the format	10	
	Total points	**50**	
Any other comments by the scrutinizer :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestionsfrom scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	**Signature of Course coordinator**
Accepted/Rejected	**Signature of Scrutinizer** **(Name:)**

Signature of HOD